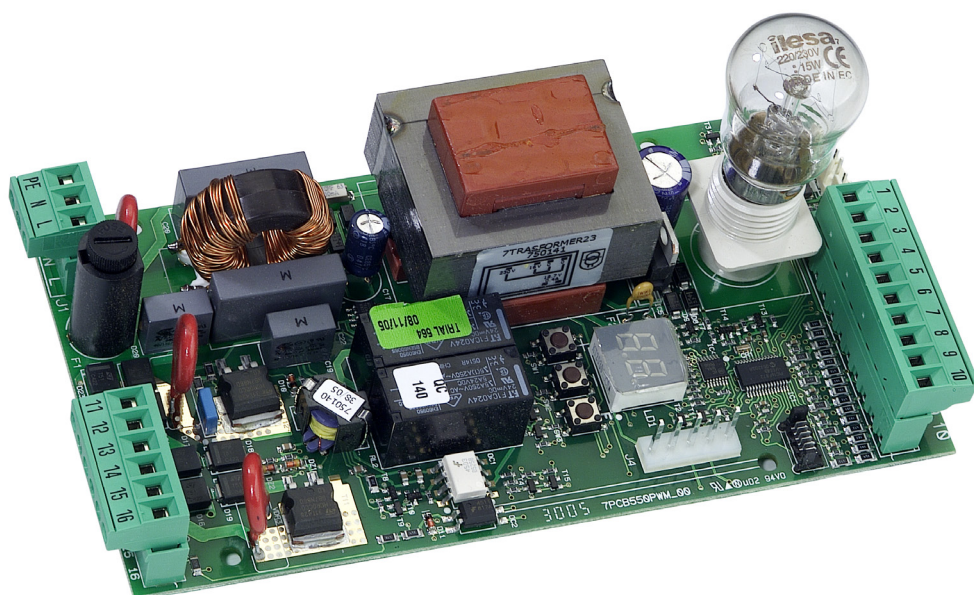


E550



FAAC

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CE DECLARATION OF CONFORMITY

Manufacturer: FAAC S.p.A.

Address: Via Calari, 10 - 40069 Zola Predosa BOLOGNA - ITALY

Declares that: Control board E550

- conforms to the essential safety requirements of the following EEC directives
2006/95/EC Low Voltage Directive
2004/108/EC Electromagnetic Compatibility Directive

Additional note:

This product has undergone testing in a typical
standard configuration (all products built by FAAC S.p.A.)

Bologna, 01-08-2013

The Managing Director
A. Marcellan



WARNINGS

- Attention! To ensure the safety of people, it is important that you read all the following instructions.
- Incorrect installation or incorrect use of the product could cause serious harm to people.
- Carefully read the instructions before beginning to install the product and keep for future reference.
- The symbol  highlights notes that are important for personal safety and the protection of the automated system.
- The symbol  calls your attention to notes on product specifications or operation.

CONTROL BOARD E550

1 WARNINGS



Attention: Before attempting any work on the control board (connections, maintenance), always turn off power.

- Install, upstream of the system, a differential thermal breaker with adequate tripping threshold,
- Connect the earth cable to the terminal on the J1 connector of the board (see fig. 1).

Always separate power cables from control and safety cables (push-button, receiver, photocells).

To avoid any electrical disturbance, use separate sheaths or a screened cable (with the screen earthed).

The **E550** control board was designed to command the opening of up-and-over doors with one or two motors.

Thanks to encoder control and opening and closing limit switches, if correctly used, it can guarantee installation in conformity with current safety regulations.

The **E550 SLAVE** control board is an interface for the second motor, on which the courtesy light is also installed.

2 TECHNICAL SPECIFICATIONS

Power supply voltage Vac (+6% -10%) / Hz.	230 / 50.
Absorbed power (W)	12
Max motors load (W)	800
Power supply for accessories (Vdc)	24
Accessories max.current (mA)	300
Operating ambient temperature (°C)	-20 / +55
Protective fuses	mains / accessories circuit
Rapid connector	Coupling to 5-pin Minidec board Decoder, Receiver, RP/RP2
Terminal board inputs (removable)	Open / Encoder / Opening safety devices / Closing safety devices / Opening limit switch / Closing limit switch / Flashing light 230 Vac - 60W.
Terminal board outputs (removable)	Motor/Outside courtesy light 230 Vac / 24 Vdc power supply to accessories
Max load for integrated courtesy light (W)	25
Max load for outside courtesy light (W)	250

4 E550 LAY-OUT AND COMPONENTS

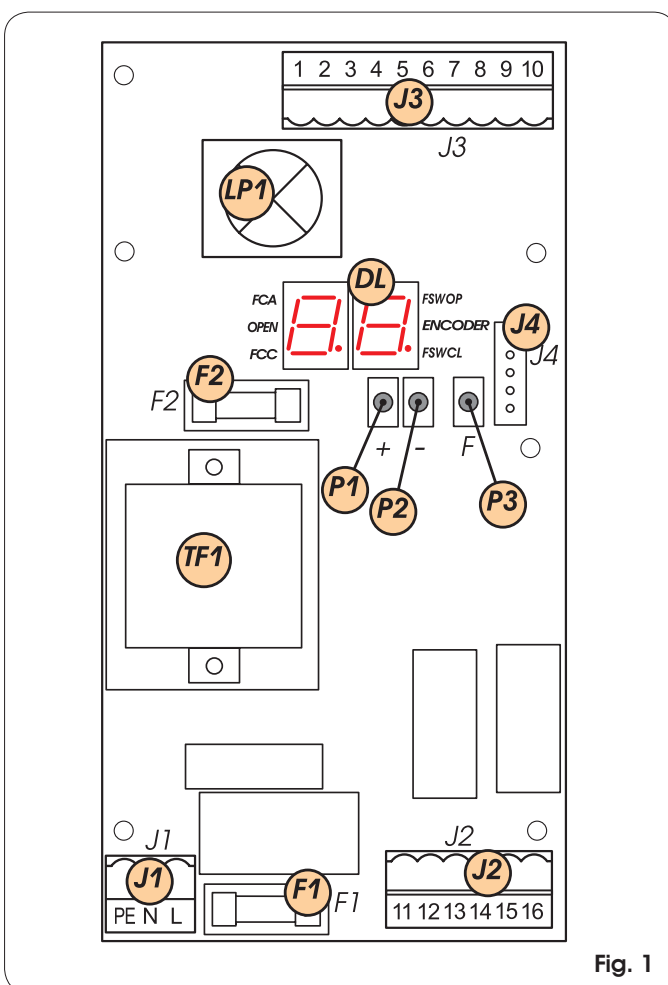


Fig. 1

3 OPERATING PARAMETERS

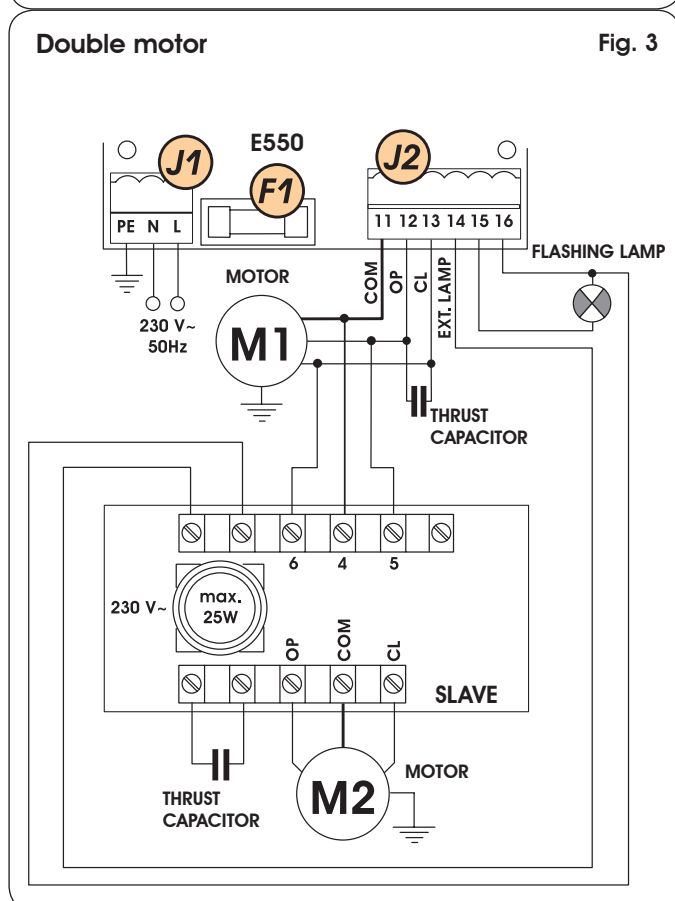
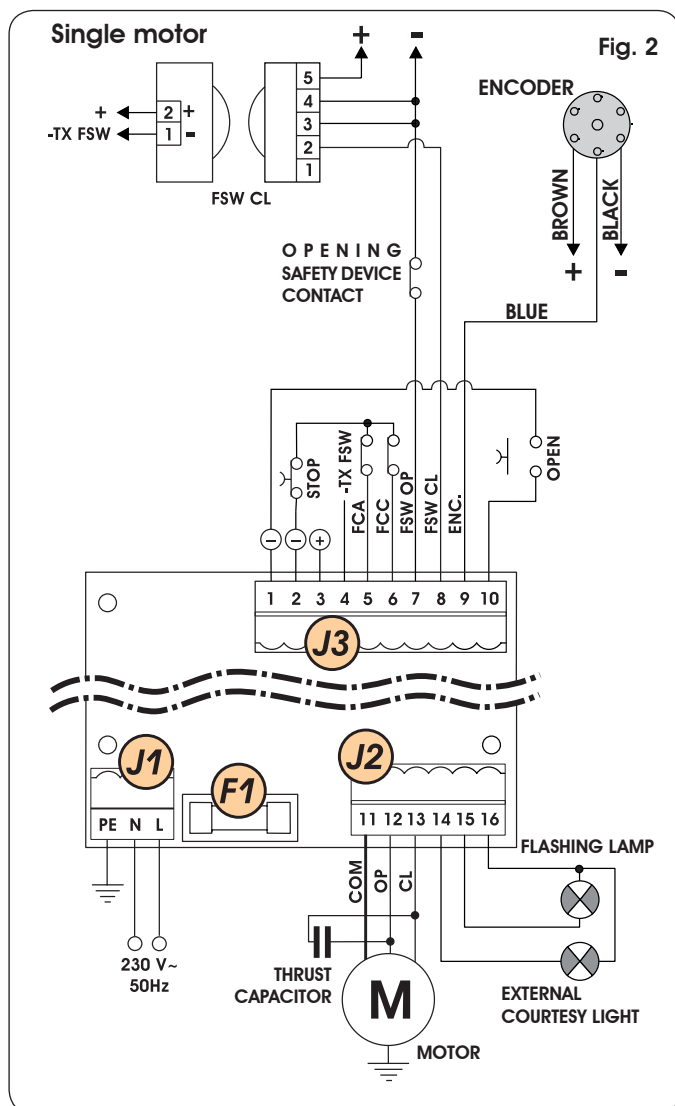
Logic	Automatic / Semi-automatic
Pause Time	Programmable from 0 to 4 min. (default 2 min.)
Max. thrust torque	YES / NO
Fail Safe	YES / NO
Pre-flashing	Programmable from 0 to 10 sec. (default 0 sec.)
Electronic clutch	Programmable on 50 levels
Limit switch intervention mode	2 types of operating modes
Courtesy light timer	Programmable from 0 to 4 min. (default 30 sec.)
Safety devices tripping mode	3 types of operating modes

4.1 DESCRIPTION OF COMPONENTS

F1	FUSE 5x20 5A/250V (mains circuit)
F2	FUSE 5x20 500mA / 250V (accessories)
J1	TERMINAL-BOARD for 230 Vac power input
J2	TERMINAL-BOARD motor output, flashing light and ext. courtesy light
J3	TERMINAL-BOARD: Low voltage inputs / accessories
J4	Rapid CONNECTOR: DECODER MINIDEC / RP RECEIVER
P1	"+" PUSH-BUTTON for programming
P2	"-" PUSH-BUTTON for programming
P3	"F" PUSH-BUTTON for programming
DL	DISPLAY for signals and programming
LP1	Courtesy LIGHT 25W 230Vac E14
TF1	TRANSFORMER

5 ELECTRICAL CONNECTIONS

5.1 DESCRIPTION OF CONNECTIONS



5.1.1 TERMINAL BOARD J1 (high voltage)

Terminal board for 230V~ 50Hz (L= Phase N=Neutral) power supply.

Connect earth of electrical system and earth cable of the operator to terminal "PE".

5.1.2 TERMINAL BOARD J2 (high voltage)

230V~ Terminal board for connection of:

Motor: connect the motor phases (Black and Brown cables) to the OP and CL terminals. Connect the common contact (Blue cable) to terminal COM. Connect the thrust condenser in parallel with respect to the phases.

Flashing lamp: connect a flashing lamp with max power of 60W between terminals 15 and 16.

Courtesy light: connect the courtesy light of the SLAVE board between terminals 14 and 16, or connect the external lights with max. overall power of 250W between these terminals.

5.1.3 TERMINAL BOARD J3 (low voltage)

1 = ⊖ = inputs common contact / Negative of accessories supply

2 = ⊖ = inputs common contact / Negative of accessories supply

3 = ⊕ = Positive for powering 24Vdc (⊕) accessories.

Max accessories load: 300mA.



To calculate absorption values, refer to the instructions for individual accessories.

4 = -FSW TX = Negative for power supply to photocell transmitters. Separate connection of the negative pole of the transmitters makes it possible to use the Failsafe control on the photocells, thus increasing the system's safety level. If the Failsafe control is disabled, the terminal is a negative for supply to accessories.

5 = FCA = Opening Limit switches contact (N.C.).

The opening limit switch consists of a micro-pushbutton which, activated by the cam when the door reaches open position, stops movement immediately or, according to programming, after 3 sec.

6 = FCC = Closing limit switches contact (N.C.).

The closing Limit switch consists of a micro-pushbutton which, activated by the cam when the door reaches closed position, stops movement immediately or, according to programming, after 3 sec.



For control unit operation, both the limit switches MUST BE connected.

7 = FSW OP = Opening Safety devices contact (N.C.)

Safety devices are all devices (photocells, sensitive edges,...) with N.C. contact, which, if there is an obstacle in the area they protect, operate by stopping or reversing the door opening movement (see chap.7.4 - Advanced programming).

They have no effect during the closing stage.

If the opening safety devices are engaged while the door is closed, they inhibit any Open pulse.

To install several safety devices, connect the N.C. contacts in series.



If no opening safety devices are connected, jumper connect "FSW OP" to terminal 4.

8 = FSW CL = Closing Safety devices contact (N.C.)

Safety devices are all devices (photocells, sensitive edges,...) with N.C. contact, which, if there is an obstacle in the area they protect, operate to reverse door closing movement. They have no effect during the opening stage, except in the Ad function (see chap.7.4. - Advanced programming).

If the closing safety devices are engaged while the door is open, they inhibit any Open pulse.

To install several safety devices, connect the N.C. contacts in series.



If no closing safety devices are connected, jumper connect "FSW CL" to terminal 4.

9 = ENCODER = Input for encoder signal.

The signal received from the encoder must be connected to this terminal. To supply power to the encoder, connect the remaining two wires to terminals "1 - 3" observing the polarity indicated on the lay-out of Fig. 2.



The control unit will not operate without an encoder.



For two-motor applications, the encoder must be applied only to motor 1 (the one with the board).

10 = OPEN = OPEN Command (N.O.)

Any device (push-button, detector,...) which, by closing a contact, supplies an opening pulse to the door.

To install several Open devices, connect N.O. contacts in parallel.

STOP = STOP command (N.C.)

A suitable device which, by opening a contact, immediately stops door movement. Connect the **STOP** in series to the common contact of the limit switches (see Fig 2) If a **STOP** Push-button cannot be connected, connect the common contact of the limit switches to terminal 2. To install several **STOP** devices, connect N.C. contacts in series.

5.1.4. CONNECTOR J4 (low voltage)

Connector J4 (Fig. 4) is used for rapid connection of Minidec, Decoder and RP Receivers. The receiver/decoding board must be inserted with the components facing the board.



Insert and remove the board **ONLY** after cutting power.

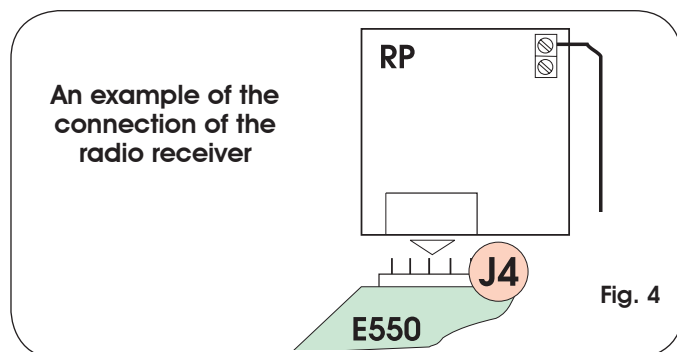
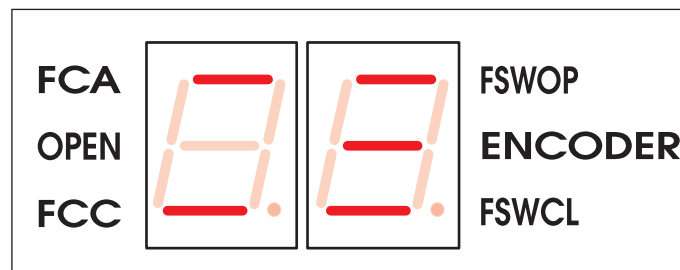


Fig. 4

6 SIGNALLING LEDS (status of inputs)

A two-digit display is installed on the board. During normal operation, it indicates the status of the inputs. We show below the exact correspondence between the horizontal segments (we will call them LEDs from now on) of the display and the inputs.

The table below indicates the status of the LEDs in relation to the status of the inputs.



Tab.1 STATUS OF LEDS

LED	ON	OFF
OPEN	Opening command active	Opening command inactive
ENCODER	LED flashes during operation	
FSW OP	Opening safety devices released	Opening safety devices engaged
FSW CL	Closing safety devices released	Closing safety devices engaged
FCA	Opening limit switch free	Opening limit switch engaged
FCC	Closing limit switch free	Closing limit switch engaged

Led on = closed contact / led off = open contact



The statuses of the LEDs with door closed and at rest are highlighted in **bold**.

7 PROGRAMMING

Power up the system, checking that no pulse generator is active and can send an **OPEN** command.

With this control board, you can adjust the force, speed and sensitivity parameters in compliance with what is specified by current regulations to respect the impact curve. (see paragraph 7.2).

To program the operation of the automated system, you must access the "PROGRAMMING" mode.

Programming is in two parts: BASIC PROGRAMMING and ADVANCED PROGRAMMING.



modification of the programming parameters is immediately effective, whereas definitive memory-storage occurs only on exiting programming and returning to the view of the status of the automated system. If you cut power to the board before returning to view the status, all the modifications made will be lost.



You can return to viewing the status from any point of programming at any level, by pressing keys F and - simultaneously.

7.1 BASIC PROGRAMMING

To access BASIC PROGRAMMING, use push-button **F**:

- if you press it (and hold it down), the display shows the name of the first function.
- if you release the push-button, the display shows the value of the function, which can be changed with keys + and -.
- if you press **F** again (and hold it down), the display shows the name of the next function.
- when you reach the last function, press the **F** push-button to exit programming, and the display resumes showing the inputs status.

The following table (Tab.2) indicates the sequence of functions accessible in BASIC PROGRAMMING:

Tab. 2

BASIC PROGRAMMING 		
Display	Function	dF01
dF	DEFAULT PROGRAMMING: 00 Neutral condition. 01 Default 01 loaded 02 Default 02 not used 03 Default 03 not used 04 Default 04 not used If you do not wish to load or modify a default, leave dF on value 00 .	00
LO	FUNCTION LOGICS: E Semi-automatic EP Semi-automatic "Step-by-step" A Automatic AP Automatic "Step-by-step"	E
PA	PAUSE TIME: This operates only if an automatic logic was selected. Can be adjusted from 0 to 59 sec in one second steps. Next, the viewing changes in minutes and tenths of a second (separated by a dot) and time is adjusted in 10 second steps, up to the maximum value of 4.1 minutes. ES: if the display shows 2.5 , the pause time will be 2 min and 50 sec.	2.0
In	To exit programming and return to display of inputs status. (see chapter 6)	

7.2 ADJUSTMENT OF OPENING AND CLOSING LIMIT SWITCHES

- 1) Release the operator.
- 2) Open the door to the required point; adjust the cam until the FCA micro push-button is activated.
- 3) Tighten the screw on the cam.
- 4) Close the door to the required point; adjust the cam until the FCC micro push-button is activated.
- 5) Tighten the screw on the cam.
- 6) Relock the operator.

7.3 ROTATION DIRECTION CHECK

Procedure for checking exact connection of motor phases:

- 1) Put the operator in manual mode.
- 2) Manually take the door to its opening mid-point.
- 3) Lock the operator.
- 4) Power up the system.
- 5) Send an opening pulse (**OPEN**) and check if the motor executes a door opening operation. If the door closes, change over the phases of the electric motor on the terminal-board (black and brown cables) of the board.

For two-operator applications: assign the same wire colours to the "**COM,OP,CL**" terminals of the E550 board and of the SLAVE board, and, if you have to change over the phases, change them over for both motors.



- The board carries out an electronic control (the connected motor must be present) before every start. If you attempt to make the board operate without motor load or with an insufficient load, no power is supplied at motor output.



- The courtesy light lights up when the motor starts and remains lighted for the programmed time, starting when the movement has finished.

7.4 ADVANCED PROGRAMMING

To access ADVANCED PROGRAMMING, use push-button **F** and +:

- 1) press push-button **F** and, while holding it down, press push-button +.
- 2) if you release the push-button +, the display shows the name of the first function (**P0**).
- 3) if you release the **F** push-button, the display shows the value of the function, which can be changed with keys + and -.
- 4) if you press **F** again (and hold it down), the display shows the name of the next function, etc.
- 5) when you reach the last function, press the **F** push-button to exit programming, and the display resumes showing the inputs status (see chapter 6).
- 6) At any point of ADVANCED PROGRAMMING, you can save the modifications and exit directly by pressing push-button **F** and, while holding it down, simultaneously pressing push-button -.

The following table (Tab.3) indicates the sequence of functions accessible in ADVANCED PROGRAMMING.

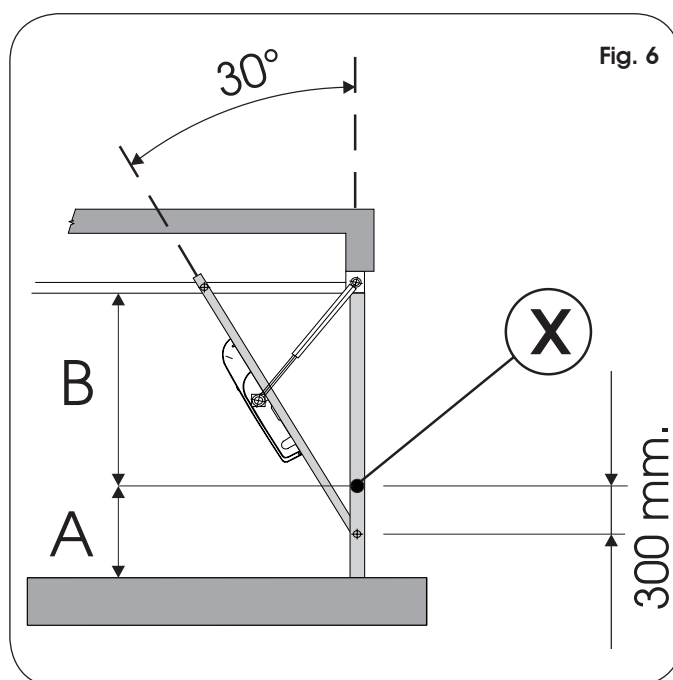
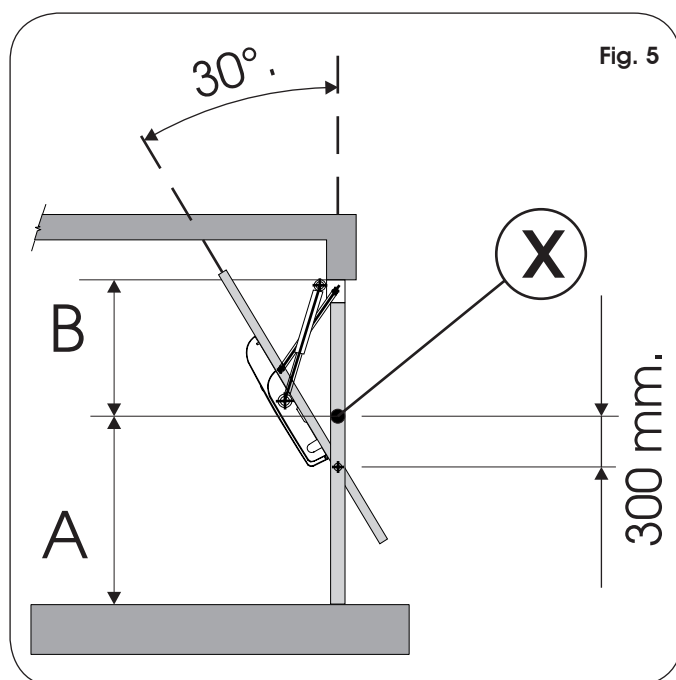
7.5 IDENTIFYING THE DOOR POSITION LIMITS

As it moves, and according to its position, the up-and-over door requires different force, speed and sensitivity. This control board is provided with independent adjustment of parameters (**F1, S1, E1, F2, S2, E2**) during closing, between the top (zone B) and bottom (zone A) of the door.

To pick out the two zones A and B, point X must be located; refer to Fig. 5 or 6.

- 1) Release the operator.
- 2) Position the up-and-over door at 30°.
- 3) To facilitate positioning the door at 30°, use the positioning template, located at the end of this manual.
- 4) Locate and mark point X as shown in Fig. 5 or 6.
- 5) Close the door by hand, relock the operator and power up the system.
- 6) Access ADVANCED PROGRAMMING with push-buttons **F** and +, and display the first function (**P0**).
- 7) Give an OPEN command, open the door electrically, inclining it until you reach the coincidence with point X.
- 8) Stop the door with another OPEN command.
- 9) Take a note of the value indicated on the display as the value of parameter **P1**, for subsequent input (e.g. **P1 = 20**).
- 10) Give an OPEN command, and open the door completely.
- 11) Wait for the motor to stop.
- 12) Take a note of the value indicated on the display as the value of parameter **P2**, for subsequent input (e.g. **P2 = 32**).
- 13) Input values **P1** and **P2** which you had annotated in the ADVANCED PROGRAMMING parameters.

Run the impact curve test and, if necessary, modify values **F1, S1, E1, F2, S2, E2** in order to respect current legal regulations.



Tab. 3

ADVANCED PROGRAMMING  + 		
Display	Function	dF01
P0	POSITION OF DOOR: Shows the position of the door during motorised movement from 00 to 59. For an explanation of parameter P0 refer to chapter 7.5.	00
P1	UPPER LIMIT OF ZONE A: Defines the upper limit of zone A, in which to subsequently adjust the force, speed and sensitivity functions of the encoder F1, S1, E1. (input value P1 as explained in chapter 7.5).	20
P2	UPPER LIMIT OF ZONE B: Defines the upper limit of zone B, in which to subsequently adjust the force, speed and sensitivity functions of the encoder F2, S2, E2. (input value P2 as explained in chapter 7.5).	32
oF	OPENING FORCE: Adjusts force during opening from 01 to 50.	25
F2	CLOSING FORCE IN ZONE B: Adjusts force during closing from 01 to 50.	30
F1	CLOSING FORCE IN ZONE A: Adjusts force during closing from 01 to 50.	20
S2	CLOSING SPEED IN ZONE B: Adjusts closing speed in zone B from 01 to 10 (10 = max. speed, 01 = min. speed).	05
S1	CLOSING SPEED IN ZONE A: Adjusts closing speed in zone A from 01 to 10 (10 = max. speed, 01 = min. speed).	03
E2	ENCODER SENSITIVITY IN ZONE B: Adjusts anti-crushing sensitivity in zone B from 01 to 50 (50 = min. sensitivity, 01 = max. sensitivity).	04
E1	ENCODER SENSITIVITY IN ZONE A: Adjusts anti-crushing sensitivity in zone A from 01 to 50 (50 = min. sensitivity, 01 = max. sensitivity).	03
LP	COURTESY LIGHT: To set lighted time of the courtesy light. Can be adjusted from 0 to 59 sec in one-second steps. Afterwards, the viewing changes in minutes and tenths of a second (separated by a dot) and time is adjusted in 10 second steps, up to the maximum value of 4.1 minutes. E.g. if the display shows 2.5, the pause time will be 2 min and 50 sec.	30

bo	MAXIMUM TORQUE AT THRUST: y = Active no = Excluded	y
FS	FAIL SAFE (safety devices operational check): y = Active no = Excluded	no
PF	PRE-FLASHING of flashing light: Can be adjusted from 0 to 10 sec in one-second steps.	00
FC	LIMIT SWITCH TRIPPING MODE: no = FCA/FCC: Stops immediately. y = FCA/FCC: Stops after a 3 sec. slow-down.	y
Ph	SAFETY DEVICES TRIPPING MODE: Sets motor operation when safety devices engaged: St = FSW OP locks opening movement, and on release, resumes opening. Ad = When up-and-over idle (closed, open or locked) and FSW CLs engaged, an OPEN pulse activates the flashing light, but movement begins only when FSW CL ⁽¹⁾ (ADMAP function) disengages. While the door is moving, FWS OP reverses and FSW CL locks and reverses at disengagement ⁽¹⁾ . CL = FSW OP reverses the opening movement, FSW CL reverses closing movement.  ⁽¹⁾ With pre-flashing selected, movement begins after the set pre-flashing time.	CL
In	To exit programming and return to display of inputs status. (see chapter 6)	

8 AUTOMATED SYSTEM TEST

When you have finished programming, check if the system is operating correctly.
Check if force is adequately adjusted and if the safety devices operate correctly.
Also run the impact test and see if it conforms to current regulations.
Otherwise, change the force, speed and sensitivity parameters in order to be again within the set values.

9 OPERATION OF ENCODER AND ELECTRONIC CLUTCH (ANTI-CRUSHING)

Thanks to encoder control and an electronic system for adjusting motor torque, the **E550** board, if correctly used, guarantees installation conforming to current safety regulations.
The encoder is active during both door opening and closing stages, until one of the two limit switches is tripped.
The encoder intervenes during opening, stopping the door movement.
During the closing stage, the encoder reverses door motion until the door is completely reopened.
The board also has a motor torque adjustment electronic system which, according to the setting, limits door thrust, respecting current legal regulations.



- Encoder intervention **DOES NOT DISABLE** automatic re-closing.
- The electronic clutch must be adjusted in conformity with current safety regulations.

10 FUNCTION LOGICS

Tab. 4/a

LOGIC "E"	PULSES			
STATUS OF THE AUTOMATED SYSTEM	OPEN	FSW CL	FSW OP	STOP
CLOSED	opens ⁽¹⁾	no effect (function Red excluded)	no effect (inhibits opening)	no effect (inhibits opening)
OPENING	stops operation	no effect	see programming	stops operation
OPEN	closes ⁽¹⁾	no effect (function Red excluded)	no effect	stops operation
CLOSING	reverses immediately at opening	reverses immediately at opening	no effect	stops operation
STOPPED	closes ⁽¹⁾	inhibits closing	no effect	no effect (inhibits opening)

Tab. 4/b

LOGIC "EP"	PULSES			
STATUS OF THE AUTOMATED SYSTEM	OPEN	FSW CL	FSW OP	STOP
CLOSED	opens ⁽¹⁾	no effect (function Red excluded)	no effect (inhibits opening)	no effect (inhibits opening)
OPENING	stops operation	no effect	see programming	stops operation
OPEN	closes ⁽¹⁾	no effect (function Red excluded)	no effect	stops operation
CLOSING	stops operation	reverses immediately at opening	no effect	stops operation
STOPPED	restarts in opposite direction (always closes after a Stop)	inhibits closing	no effect	no effect (inhibits opening)

Tab. 4/c

LOGIC "A"	PULSES			
STATUS OF THE AUTOMATED SYSTEM	OPEN	FSW CL	FSW OP	STOP
CLOSED	opens and closes again after pause time ⁽¹⁾	no effect (function Red excluded)	no effect (inhibits opening)	no effect (inhibits opening)
OPENING	no effect	no effect	see programming	stops operation
OPEN IN PAUSE	recharges pause time	no effect (inhibits closing)	no effect	stops operation
CLOSING	reverses immediately at opening	reverses immediately at opening	no effect	stops operation
STOPPED	closes ⁽¹⁾	no effect (inhibits closing)	no effect	no effect (inhibits opening)

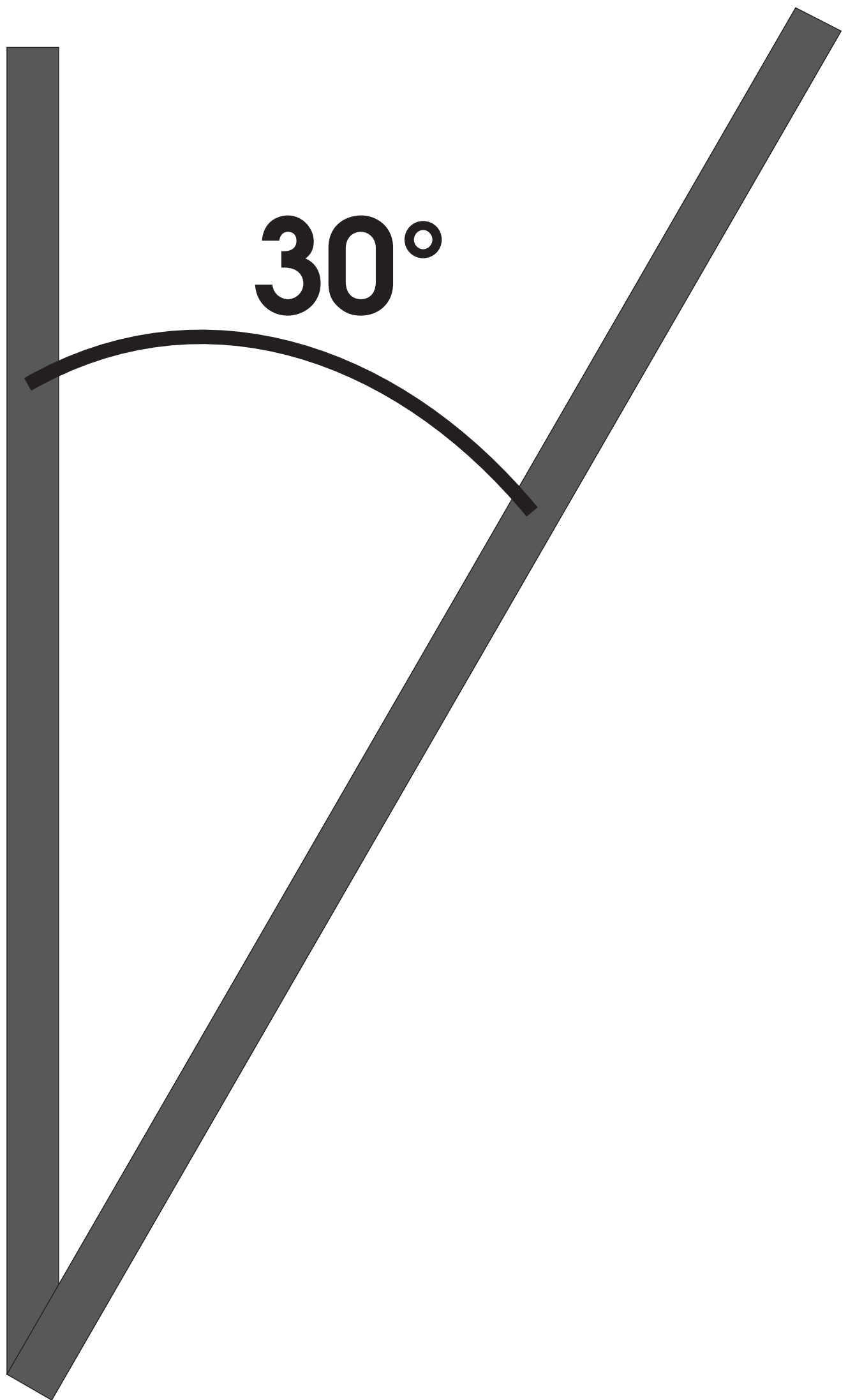
Tab. 4/d

LOGIC "AP"	PULSES			
STATUS OF THE AUTOMATED SYSTEM	OPEN	FSW CL	FSW OP	STOP
CLOSED	opens and closes again after pause time ⁽¹⁾	no effect (function Red excluded)	nessun effetto (inibisce apertura)	no effect (inhibits opening)
OPENING	stops operation	no effect	see programming	stops operation
OPEN IN PAUSE	stops operation	no effect (inhibits closing)	no effect	stops operation
CLOSING	reverses immediately at opening	reverses immediately at opening	no effect	stops operation
STOPPED	closes ⁽¹⁾	no effect (inhibits closing)	no effect	no effect (inhibits opening)

⁽¹⁾ With pre-flashing selected, movement begins after the set pre-flashing time.



When the Encoder is tripped, this causes stop at opening and reverse at closing.



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